AALBORG UNIVERSITY
Department of Computer Science

Software Engineering for AI Systems
Spring 2027

Portfolio 3: Big Data & Data Science Technologies

Case: Short Anonymised Case Name

Student Name

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Assessment criteria (delete this box before submitting):

- The volume estimate is calculated and drives the processing and storage choices.
- Governance rules are concrete and consistent with the P2 data-placement table.
- Drift and quality signals are defined, measurable and tied to an action.

1 Data Flow

Show how data flows from source to storage to model, with the tier boundaries from P2.

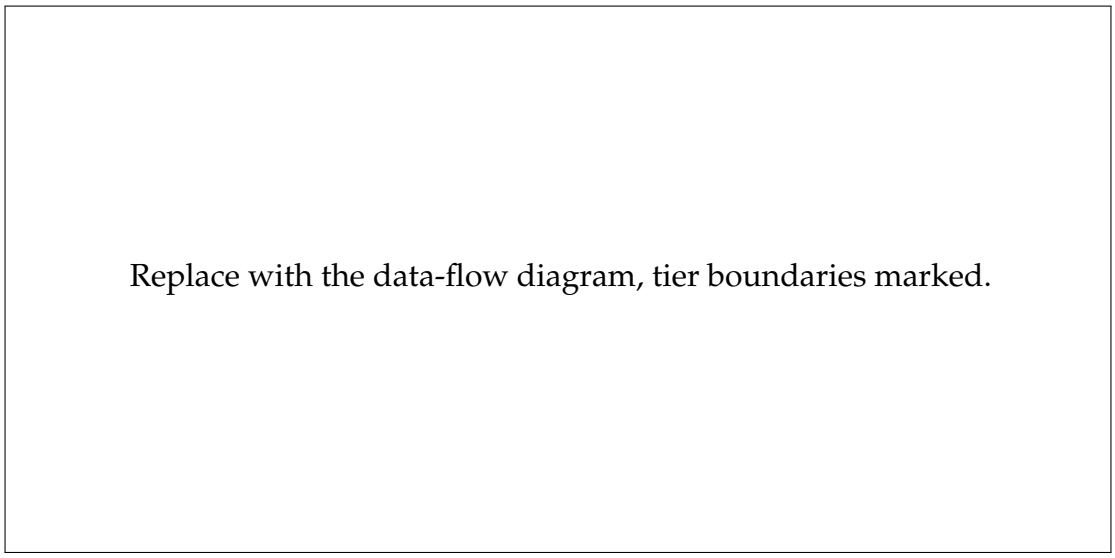


Figure 1: Data flow.

2 Volume and Rate

Estimate what data the system produces and consumes, at what volume and rate. Show the calculation, e.g.

$$\text{volume per day} = \text{sources} \times \text{records per second} \times \text{bytes per record} \times 86\,400.$$

Data source	Rate	Size per record	Volume per day
…	…	…	…
…	…	…	…

Table 1: Volume and rate estimate.

3 Processing and Storage

Choose batch, stream or both, and where historical and event data are stored (topic retention, data lake, warehouse, feature store) [1]. Use the estimate in Table 1 as an argument.

3.1 ADR-4: Processing

Status: Proposed / accepted.

Context: The volume, rate and latency that drive the choice.

Alternatives: Batch, stream, or both.

Decision: The chosen alternative.

Consequences: The trade-off that decided it, and what gets harder.

3.2 ADR-5: Storage

Status: ...

Context: ...

Alternatives: ...

Decision: ...

Consequences: ...

4 Data Governance

State who owns which data, how long it is kept, and which regulations apply (e.g. GDPR, NIS2, EU AI Act), and what they require of the design. The locations must match the P2 data-placement table.

Data item	Owner	Retention	Location	Applicable rule
...	...	...	...	...
...	...	...	...	...

Table 2: Data governance.

5 Observability

5.1 Data Quality

Describe the checks on incoming data (e.g. schema, ranges, missing values) and where they run.

5.2 Drift and Proxy Metrics

Describe how drift is detected while labels arrive late, and which proxy metrics stand in for model quality until they do.

5.3 Service and Model Quality

Separate service SLOs from model-quality metrics.

Metric	Kind	Target	Measured where
…	Service SLO	…	…
…	Model quality	…	…
…	Data quality / drift	…	…

Table 3: Observability metrics.

5.4 Feedback Loop

Tie each signal to an action: retrain, roll back or reconfigure.

Signal	Threshold	Action
…	…	…

Table 4: Signals and actions.

6 Evidence

State whether the evidence comes from your own case or from the reference project, and show the running pipeline. Dashboards and drift evidence from week 6 may be added in the final report.

7 AI Usage

Briefly describe your use of AI tools on this assignment:

- *Code generation:* …
- *Grammar and writing:* …
- *Clarification of technical aspects:* …

Bibliography

[1] C. Huyen, *Designing Machine Learning Systems*. Sebastopol, CA: O'Reilly Media, 2022.